CSE400 Joint Random Variables

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Assume you are expert lecture scribe generator. Generate lecture scribe suitable for exam preparation for CSE400 Joint Random Variables + Joint Distributions + Examples
While writing the scribe, explain everything step by step so that the logic and reasoning are clear. The scribe should include all important things like definitions, assumptions, concepts, properties, derivations and worked examples mentioned in the lecture. It should master key of revision.
Use only the provided lecture slides. Do not add any explanations, examples or anything that is not present in the given context. Do not use any outside sources.
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Make it like the scribe covers every important point from the slides and explains the steps clearly so that it is easy to revise before exams.

Below is a **clean exam-revision lecture scribe** created **strictly from the provided slides**. I did **not add outside concepts** and kept the explanations aligned with the lecture flow so it works as a **revision master document**.

CSE400 – Lecture Scribe

Joint Random Variables + Joint Distributions + Examples

(Based only on provided lecture slides)

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1. Introduction: Why Study Joint Random Variables?

In many real-world systems, **more than one random variable (RV)** is involved simultaneously. Studying a single random variable is often not enough because the

variables may **depend on each other** or **influence each other**.

Therefore, we study **Joint Random Variables**, which allow us to analyze the behavior of **two or more random variables together**.

Motivation Through Examples

Example 1: Smart Transportation System

Two random variables are considered:

- X = Vehicle Speed
- Y = Traffic Density

Key question:

> Can vehicle speed be high when traffic density is also high?

The answer depends on the relationship between the variables. In many situations:

- Higher density usually means lower speed.
- However, this relationship is not deterministic.

Thus:

X and Y are correlated random variables.

Applications in Intelligent Transportation Systems (ITS):

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Key question:

> If SNR is high, will throughput always be high?

Answer: **Not always.**

Reason:

Even when SNR is high, throughput can still be affected by:

- Network congestion
- Interference
- Fading
- Scheduling

Applications in wireless networks:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction

Random variables:

- X = Rainfall
- Y = Temperature

These variables are jointly analyzed for **weather forecasting**.

Example 4: Drone Delivery

Random variables:

- X = Wind Speed
- Y = Delivery Time

Higher wind speeds can influence delivery time in **UAV and logistics systems**.

Example 5: Rolling Two Dice

Let:

- X = result of first die
- Y = result of second die

We may want to compute probabilities such as:

- $X + Y = 7$
- $X > Y$

To solve such problems we require the **Joint Probability Mass Function (Joint PMF)**.

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2. Discrete Case – Joint PMF

When the random variables are **discrete**, their joint behavior is described using the **Joint Probability Mass Function (Joint PMF)**.

The Joint PMF specifies the probability that:

$$X = x \quad \text{and} \quad Y = y$$

simultaneously.

Thus,

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This function gives the probability for **every possible pair of values** of X and Y.

Joint PMF Table Representation

A convenient way to represent a joint PMF is using a **table**.

Structure of the table:

Y / X	x_1	x_2	x_3	...
y_1	$p(x_1,y_1)$	$p(x_2,y_1)$	...	
y_2	$p(x_1,y_2)$	$p(x_2,y_2)$	...	

Each cell represents:

$$P(X = x, Y = y)$$

Key idea:

The table contains **probabilities of all combinations of values**.

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3. Example: Rolling Two Dice

Consider two dice being rolled.

Define:

- X = outcome of first die
- Y = outcome of second die

Each die can take values:

$$1, 2, 3, 4, 5, 6$$

Thus there are:

$$6 \times 6 = 36$$

possible outcomes.

Each outcome corresponds to a pair:

$$(X, Y)$$

Examples:

- (1,1)
- (1,2)
- (2,3)
- (6,6)

Since the dice are fair:

Each pair has probability:

$$\frac{1}{36}$$

The joint PMF table therefore contains probabilities for all **36 combinations**.

This table allows us to answer questions like:

- $P(X + Y = 7)$
- $P(X > Y)$

Such probabilities cannot be computed using single-variable distributions alone.

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4. Continuous Case – Joint PDF

When random variables are **continuous**, we use the **Joint Probability Density Function (Joint PDF)**.

The joint PDF is written as:

$$f_{X,Y}(x, y)$$

Interpretation:

$$f_{X,Y}(x, y)\, dx\, dy$$

represents the probability that:

- X lies in the interval x to $x + dx$
- Y lies in the interval y to $y + dy$

Geometric Interpretation

In the continuous case:

- The joint PDF can be visualized as a **surface over the x-y plane**.
- Probability corresponds to the **volume under this surface**.

Thus:

Probability of a region R is computed by integrating:

$$P((X, Y) \in R)$$

over that region.

---

5. Worked Example (Continuous Case)

A joint PDF is given over a certain region.

The steps used to solve the example are:

1. Identify the **region where the PDF is defined**.

2. Write the required probability expression.
3. Set up the **double integral** over the region.
4. Integrate step by step.
5. Obtain the final probability.

The example demonstrates how probabilities of continuous joint variables are obtained through **integration over the valid region**.

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6. Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** describes cumulative probability for two random variables.

Definition:

$$F_{X,Y}(x, y) = P(X \leq x,\ Y \leq y)$$

Meaning:

It gives the probability that:

- X is less than or equal to x
- and Y is less than or equal to y

at the same time.

---

7. Joint CDF – Continuous Case

For continuous variables, the joint CDF is obtained from the joint PDF.

The relation is:

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, du\, dv$$

Interpretation:

- Integrate the joint PDF from negative infinity up to x and y.
- This accumulates the probability up to that point.

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8. Joint CDF – Discrete Case

For discrete random variables:

The joint CDF is computed by **summing probabilities**.

$$F_{X,Y}(x,y) = \sum_{u \le x} \sum_{v \le y} P(X = u, Y = v)$$

Thus we add the probabilities of all pairs satisfying:

$$X \le x, \quad Y \le y$$

---

9. Properties of the Joint CDF

The joint cumulative distribution function has several properties.

1. Range

$$0 \le F_{X,Y}(x,y) \le 1$$

Because it represents probability.

---

2. Monotonicity

The joint CDF is **non-decreasing** in both variables.

Meaning:

- Increasing x cannot decrease the probability.
- Increasing y cannot decrease the probability.

---

3. Limits

As x and y go to infinity:

$$F_{X,Y}(x,y) \to 1$$

because all probability mass is accumulated.

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10. Key Exam Revision Summary

Why Joint Random Variables?

Used when analyzing **multiple random variables simultaneously**.

Examples:

- Traffic systems
- Wireless networks
- Weather forecasting
- Drone delivery
- Rolling two dice

Discrete Case

Use **Joint PMF**

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Represented using **joint probability tables**.

Continuous Case

Use **Joint PDF**

$$f_{X,Y}(x, y)$$

Probability obtained by **double integration over regions**.

Joint CDF

Definition:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

Continuous case → double integral
Discrete case → summation

✔ This scribe contains the **complete conceptual flow of the slides** and is structured for **quick exam revision**.

If you want, I can also:

- **Convert this directly into LaTeX exam notes**
- **Generate a clean PDF version for studying**
- **Add slide-wise structure (Slide 1 → Slide 24 mapping)** which many CSE400 students use for revision.

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Below is a **clear, exam-revision style lecture scribe** created **strictly from the provided slides**. Nothing outside the slides has been added; the content is only reorganized and explained step-by-step so the reasoning flow is easy for revision.

CSE400 – Lecture Scribe

Joint Random Variables + Joint Distributions + Examples

Course: Fundamentals of Probability in Computing

Instructor: Dhaval Patel, PhD

Lecture: L15–L18

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1. Outline of the Lecture

The lecture covers the following topics:

1. Why study joint random variables
2. Discrete case
   - Joint PMF
   - Joint PMF table representation
   - Example: Rolling two dice
3. Continuous case
   - Joint PDF
   - Geometric interpretation
   - Worked example
4. Joint cumulative distribution function (Joint CDF)
   - Definition
   - Continuous case
   - Discrete case
   - Properties of the joint CDF
5. Comprehensive example

---

2. Why Study Joint Random Variables?

In many real systems, **multiple random variables occur at the same time**. Instead of analyzing them individually, we often need to understand **their relationship and combined behavior**.

Thus, we study **Joint Random Variables**, which allow analysis of **two or more random variables simultaneously**.

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3. Motivating Examples

Example 1: Smart Transportation

Consider two random variables:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question:

> Can speed be high when traffic density is also high?

This question studies the **relationship between the two variables**.

Thus:

- X and Y are **correlated random variables**.

Applications in Intelligent Transportation Systems:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two important random variables:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Key reasoning:

- Throughput depends on SNR.
- However, both vary due to:
  - fading
  - interference
  - congestion

Question:

> If SNR is high, will throughput always be high?

Answer: **Not necessarily**, because network congestion can influence throughput.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction

Random variables:

- $X =$ Rainfall
- $Y =$ Temperature

Application:

- Weather forecasting

Example 4: Drone Delivery

Random variables:

- $X =$ Wind Speed
- $Y =$ Delivery Time

Application:

- Logistics and UAV systems

Example 5: Rolling Two Dice

Let:

- X = value of first die
- Y = value of second die

Example questions:

- $X + Y = 7$
- $X > Y$

Such questions require the **joint probability distribution**, specifically the **joint PMF**.

---

4. Discrete Case – Joint PMF

When the random variables are **discrete**, their joint behavior is described using the **Joint Probability Mass Function (Joint PMF)**.

The joint PMF represents the probability that **both variables take specific values simultaneously**.

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This value gives the probability that:

- random variable X equals x, and
- random variable Y equals y

at the same time.

---

5. Joint PMF Table Representation

A convenient way to represent the joint PMF is through a **table format**.

Structure:

Y \ X	x_1	x_2	x_3	...
y_1	$P(x_1,y_1)$	$P(x_2,y_1)$	...	
y_2	$P(x_1,y_2)$	$P(x_2,y_2)$	...	

Each cell contains the probability:

$$P(X = x, Y = y)$$

Thus the table lists **probabilities for every possible pair of values**.

---

6. Example: Rolling Two Dice

Consider rolling two dice.

Define:

- X = result of first die

- Y = result of second die

Each die can take values:

$$1, 2, 3, 4, 5, 6$$

Thus the total number of possible outcomes is:

$$6 \times 6 = 36$$

Each outcome is a pair:

$$(X, Y)$$

Examples of possible pairs:

- (1,1)
- (1,2)
- (2,5)
- (6,6)

For fair dice:

Each pair has probability

$$\frac{1}{36}$$

Using this joint PMF table, we can compute probabilities of events such as:

- $X + Y = 7$
- $X > Y$

---

7. Continuous Case – Joint PDF

When random variables are **continuous**, we use the **Joint Probability Density Function (Joint PDF)**.

It is written as:

$$f_{X,Y}(x, y)$$

This function represents the density of probability for the pair (X, Y).

---

8. Geometric Interpretation

For continuous variables:

- The joint PDF can be visualized as a **surface over the x-y plane**.
- The probability corresponds to the **volume under the surface** over a given region.

Thus, probability for a region is obtained by **integrating the joint PDF over that region**.

---

9. Worked Example (Continuous Case)

In the lecture example, the following steps are followed:

1. Identify the region where the joint PDF is defined.
2. Express the probability that needs to be calculated.
3. Set up the **double integral** over the region.
4. Integrate step by step.
5. Obtain the required probability.

This demonstrates how probabilities are computed for **continuous joint random variables**.

---

10. Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** describes the cumulative probability for two random variables.

Definition:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

This represents the probability that:

- X is less than or equal to x, and
- Y is less than or equal to y.

---

11. Joint CDF – Continuous Case

For continuous random variables, the joint CDF is obtained by integrating the joint PDF.

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, du\, dv$$

This accumulates the probability from negative infinity up to x and y.

---

12. Joint CDF – Discrete Case

For discrete random variables, the joint CDF is obtained by summing the joint PMF values.

$$F_{X,Y}(x,y) = \sum_{u \le x} \sum_{v \le y} P(X = u, Y = v)$$

This means we add the probabilities of all pairs satisfying:

$$X \le x \quad \text{and} \quad Y \le y$$

---

13. Properties of the Joint CDF

The joint cumulative distribution function has several important properties.

1. Range

$$0 \le F_{X,Y}(x,y) \le 1$$

Since it represents probability.

---

2. Non-decreasing behavior

The joint CDF **does not decrease** as x or y increases.

---

3. Limit behavior

As x and y approach infinity:

$$F_{X,Y}(x,y) \to 1$$

because all probability mass is accumulated.

---

14. Key Points for Exam Revision

Why Joint Random Variables?

Used to analyze **two or more random variables together**.

Discrete Case

Joint PMF:

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Represented using **joint PMF tables**.

Continuous Case

Joint PDF:

$$f_{X,Y}(x, y)$$

Probabilities computed using **double integrals**.

Joint CDF

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

- Continuous case → integration
- Discrete case → summation

✅ This version is structured as a **clean exam-revision scribe**, keeping the **exact lecture content but organized logically for studying**.

If you want, I can also create a **much better "exam-killer version" of this scribe (what most CSE400 toppers use)** that includes:

- slide-by-slide explanation
- clearer derivation steps
- joint PMF dice table shown fully
- step-by-step solved example reconstruction.

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CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Joint Random Variables + Joint Distributions + Examples

Instructor: Dhaval Patel, PhD

Lecture: L15–L18

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1. Lecture Outline

The lecture is organized into the following topics:

1. Why Study Joint Random Variables
2. Discrete Case
   - Joint PMF
   - Joint PMF Table Representation
   - Example: Rolling Two Dice
3. Continuous Case
   - Joint PDF
   - Geometric Interpretation
   - Worked Example
4. Joint Cumulative Distribution Function (Joint CDF)
   - Definition
   - Continuous Case
   - Discrete Case
   - Properties of Joint CDF
5. Comprehensive Example

---

2. Why Study Joint Random Variables?

In many real-world systems, we are interested in **more than one random variable at the same time**. Studying a single random variable alone may not capture the **relationship between different quantities**.

Therefore, we study **joint random variables**, which allow us to analyze the behavior of **two or more random variables simultaneously**.

---

3. Motivating Examples

Example 1: Smart Transportation

Two random variables are defined:

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question:

> Can speed be high when traffic density is also high?

This question investigates the **relationship between these two variables**.

Thus:

- X and Y are **correlated random variables**.

Applications in Intelligent Transportation Systems include:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Two important random variables:

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Reason for studying them jointly:

- Throughput depends on SNR.
- However, both vary because of:
  - fading
  - interference
  - congestion

Question:

> If SNR is high, will throughput always be high?

Answer: Not necessarily.

Reasoning:

- Network congestion may affect throughput even if SNR is high.

Applications:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction

Random variables:

- $X =$ Rainfall
- $Y =$ Temperature

Application:

- Weather forecasting

Example 4: Drone Delivery

Random variables:

- $X =$ Wind Speed
- $Y =$ Delivery Time

Application:

- Logistics and UAV systems

Example 5: Rolling Two Dice

Let:

- X = value of first die
- Y = value of second die

Possible questions:

- $X + Y = 7$
- $X > Y$

Such problems require the **joint probability distribution**, specifically the **joint PMF**.

---

4. Discrete Case – Joint PMF

When the random variables are **discrete**, their joint behavior is described by the **Joint Probability Mass Function (Joint PMF)**.

Definition:

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This value represents the probability that:

- X takes value x, and
- Y takes value y

simultaneously.

---

5. Joint PMF Table Representation

The joint PMF is commonly represented using a **table format**.

Example structure:

Y \ X	x_1	x_2	x_3
y_1	$P(x_1,y_1)$	$P(x_2,y_1)$	$P(x_3,y_1)$
y_2	$P(x_1,y_2)$	$P(x_2,y_2)$	$P(x_3,y_2)$

Each cell in the table represents:

$$P(X = x, Y = y)$$

This table shows probabilities for **every possible pair of values** of X and Y.

---

6. Example: Rolling Two Dice

Consider rolling two dice.

Define:

- $X =$ result of the first die
- $Y =$ result of the second die

Each die can take the values:

$$1, 2, 3, 4, 5, 6$$

Total possible outcomes:

$$6 \times 6 = 36$$

Each outcome corresponds to a pair:

$$(X, Y)$$

Examples of possible outcomes:

- (1,1)
- (1,2)
- (2,4)
- (6,6)

If the dice are fair:

Each pair has probability

$$\frac{1}{36}$$

The joint PMF table lists probabilities for all **36 combinations**.

Using this table, we can compute probabilities such as:

- $P(X + Y = 7)$
- $P(X > Y)$

---

7. Continuous Case – Joint PDF

When random variables are **continuous**, we use the **Joint Probability Density Function (Joint PDF)**.

It is denoted as:

$$f_{X,Y}(x, y)$$

The joint PDF represents the probability density for the pair (X, Y).

---

8. Geometric Interpretation

For continuous random variables:

- The joint PDF can be visualized as a **surface above the x-y plane**.
- The probability corresponds to the **volume under the surface** over a region.

Therefore, probability over a region is computed by **integrating the joint PDF over that region**.

9. Worked Example (Continuous Case)

The lecture demonstrates the process of solving a problem using the joint PDF.

Steps followed:

1. Identify the region where the joint PDF is defined.
2. Write the probability expression.
3. Set up the **double integral over the region**.
4. Evaluate the integral step by step.
5. Obtain the required probability value.

This example illustrates how probabilities are computed using **integration of the joint PDF**.

---

10. Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** describes the cumulative probability for two random variables.

Definition:

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

This represents the probability that:

- X is less than or equal to x, and
- Y is less than or equal to y.

---

11. Joint CDF – Continuous Case

For continuous random variables, the joint CDF is obtained from the joint PDF.

$$F_{X,Y}(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, du\, dv$$

This integral accumulates probability from negative infinity up to x and y.

---

12. Joint CDF – Discrete Case

For discrete random variables, the joint CDF is obtained by summation.

$$F_{X,Y}(x,y) = \sum_{u \le x} \sum_{v \le y} P(X = u, Y = v)$$

Thus, we add probabilities for all pairs satisfying:

$$X \le x \quad \text{and} \quad Y \le y$$

---

13. Properties of the Joint CDF

The joint CDF satisfies several properties.

1. Range

$$0 \le F_{X,Y}(x,y) \le 1$$

Because it represents probability.

---

2. Monotonicity

The joint CDF is **non-decreasing** in both variables.

- Increasing x cannot decrease the value.
- Increasing y cannot decrease the value.

---

3. Limit Behavior

As x and y approach infinity:

$$F_{X,Y}(x,y) \to 1$$

since all probability mass is accumulated.

---

14. Key Revision Summary

Joint Random Variables

Used when studying **two or more random variables simultaneously**.

Discrete Case

Joint PMF:

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Represented using **joint PMF tables**.

Continuous Case

Joint PDF:

$$f_{X,Y}(x, y)$$

Probability computed using **double integration**.

Joint CDF

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

- Continuous → integration
- Discrete → summation

✔ This scribe reorganizes the lecture slides into a **clear step-by-step revision format** while **keeping the content strictly limited to what appears in the slides**.

If you want, I can also **turn this into a clean LaTeX-ready exam document (the format professors like for CSE400 submissions)** so you can directly compile it into a **high-quality PDF for studying.**